

CM0246 Discrete Structures

Equivalence Relations

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Preliminaries

Textbook

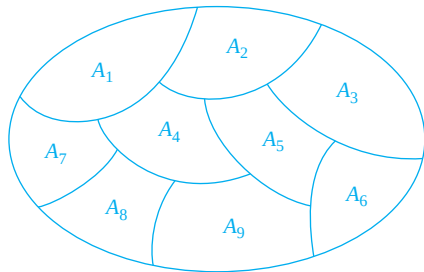
Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Introduction

Equivalence relations split sets into disjoint classes of equivalent elements.[†]



[†]Figure source: (Rosen [1988] 2012, § 9.5, Fig. 1).

Equivalence Relations

Definition

A relation on a set A is an **equivalence relation** iff it is reflexive, symmetric and transitive.

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Example (words of the same length)

$$\Sigma = \{a, b, \dots, z\},$$

$$\Sigma^* = \{w \mid w \text{ is a word on } \Sigma\},$$

$$R = \{(w, w') \mid l(w) = l(w')\} \subseteq \Sigma^* \times \Sigma^*.$$

Equivalence Relations

Exercise

Let $A = \{a, e, i, o, u\}$. Is the equality relation on A an equivalence relation?

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Example

$$\text{FUN} = \{ f \mid f : \{0, 1\} \rightarrow \{0, 1\} \},$$

$$R = \{ (f, g) \mid f(1) = g(1) \} \subseteq \text{FUN} \times \text{FUN}.$$

Equivalence Relations

Exercise

Let A be a unitary set. It is possible to define an equivalence relation on A ?

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Exercise

Define an equivalence relation on a finite/infinite set.

Equivalence Relations

Definition

Let m and n be integers and let d be a positive integer. The number m **is congruent to** n **modulo** d , denoted by $m \equiv n \pmod{d}$, iff $d \mid (m - n)$.

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Example

The congruence relation is an equivalence relation.

Equivalence Classes

Definition

Let R be an equivalence relation on a set A . The **equivalence class** of $a \in A$ with respect to R is defined by

$$[a]_R = \{ s \in A \mid (a, s) \in R \}.$$

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Example (Words of the same length)

$$[\lambda] = \{\lambda\},$$

$$[a] = \{a, b, \dots, z\} = [k],$$

$$[aa] = \{aa, ab, \dots, az, ba, bb, \dots, bz, \dots, za, \dots, zz\},$$

$$[hgbj] = \{ w \mid l(w) = 4 \}.$$

Equivalence Classes

Example (equality relation)

Whiteboard.

Equivalence Classes

Example (equality relation)

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Example (Cartesian product)

Whiteboard.

Equivalence Relations

Theorem

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Proof.

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Proof.

i) $aRb \Rightarrow [a] \subseteq [b]$

- | | | |
|---|--------------------------------|----------------------|
| 1 | $aRb.$ | (hypothesis) |
| 2 | Let $c \in [a]$. | |
| 3 | $aRc.$ | (def. of $[a]$) |
| 4 | $bRa.$ | (R is symmetric) |
| 5 | $bRc.$ | (R is transitive) |
| 6 | $c \in [b].$ | (def. of $[b]$) |
| 7 | Therefore $[a] \subseteq [b].$ | ■ |

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ii) $aRb \Rightarrow [b] \subseteq [a]$

Equivalence Relations

Theorem

Let R be an equivalence relation on a set A . For all $a, b \in A$,

$$aRb \Rightarrow [a] = [b].$$

Proof.

ii) $aRb \Rightarrow [b] \subseteq [a]$

- | | | |
|---|--------------------------------|----------------------|
| 1 | $aRb.$ | (hypothesis) |
| 2 | Let $c \in [b].$ | |
| 3 | $bRc.$ | (def. of $[b]$) |
| 4 | $aRc.$ | (R is transitive) |
| 5 | $c \in [a].$ | (def. of $[a]$) |
| 6 | Therefore $[b] \subseteq [a].$ | ■ |

Equivalence Relations

Theorem

Let R be an equivalence relation on a set A . For all $a, b \in A$,

$$[a] = [b] \Rightarrow [a] \cap [b] \neq \emptyset.$$

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Theorem

Let R be an equivalence relation on a set A . For all $a, b \in A$,

$$[a] = [b] \Rightarrow [a] \cap [b] \neq \emptyset.$$

Proof.

- 1 $[a] = [b]$. (hypothesis)
- 2 $[a] = \{a, \dots\}$. (R is reflexive)
- 3 Therefore $[a] \cap [b] \neq \emptyset$. ■

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$$[a] \cap [b] \neq \emptyset \Rightarrow aRb.$$

Proof.

- 1 $[a] \cap [b] \neq \emptyset.$ (hypothesis)
- 2 Let c such that $c \in [a]$ and $c \in [b]$.
- 3 aRc y $bRc.$ (def. of $[a]$ and $[b]$)
- 4 $cRb.$ (R is symmetric)
- 5 Therefore $aRb.$ (R is transitive)



Equivalence Relations

Theorem 1 (p. 476)

Let R be an equivalence relation on a set A . For all $a, b \in A$, the following statements are equivalent:

$$aRb, \tag{1}$$

$$[a] = [b], \tag{2}$$

$$[a] \cap [b] \neq \emptyset. \tag{3}$$

Proof.

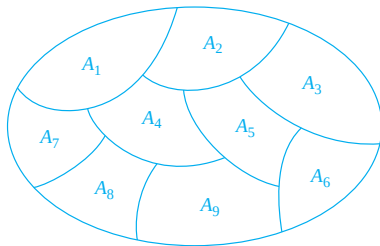
- (1) $\Rightarrow$ (2) (previous theorem),
- (2) $\Rightarrow$ (3) (previous theorem) and
- (3) $\Rightarrow$ (1) (previous theorem). ■

Equivalence Relations and Partitions

Definition

A **partition** of a set A is a collection of subsets $\{A_i \mid i \in I\}$ of A such that:[†]

- i) $A_i \neq \emptyset$, for $i \in I$,
- ii) $A_i \cap A_j = \emptyset$ when $i \neq j$ (disjoint subsets) and
- iii) $\bigcup_{i \in I} A_i = A$.



[†]Figure source: (Rosen [1988] 2012, § 9.5, Fig. 1).

Equivalence Relations and Partitions

Theorem 2 (p. 477)

Let R be an equivalence relation on a set A . Then the equivalence classes of R form a partition of A .

Proof.

The collection of subsets is given by

$$\left\{ A_{[a]_R} \mid [a]_R \text{ is an equivalence class respect to } R \right\}.$$

Using the above collection, the conditions i), ii) and iii) are satisfied. ■

Equivalence Relations and Partitions

Theorem 2 (Rosen (5th ed.), p. 477)

Given a partition $\{A_i \mid i \in I\}$ of a set A , there is an equivalence relation R that has the sets A_i as its equivalence classes.

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Given a partition $\{A_i \mid i \in I\}$ of a set A , there is an equivalence relation R that has the sets A_i as its equivalence classes.

Example

Given a partition to build the equivalence relation associated.

Equivalence Relations and Partitions

Proof.

Let R be the relation defined by

$$R = \{ (a, b) \mid a, b \in A_i \}.$$

R is a relation of equivalence:

Equivalence Relations and Partitions

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$$R = \{ (a, b) \mid a, b \in A_i \}.$$

R is a relation of equivalence:

- Reflexivity and symmetry

Direct from the definition of R .

(continued on next slide)

Equivalence Relations and Partitions

Proof (continuation).

$$R = \{ (a, b) \mid a, b \in A_i \}.$$

- Transitivity

- 1) aRb and bRc .
- 2) Exists $X \in \{ A_i \mid i \in I \}$ such that $a, b \in X$ by definition of R .
- 3) Exists $Y \in \{ A_i \mid i \in I \}$ such that $b, c \in Y$ by definition of R .
- 4) $X = Y$ because $b \in X$ and $b \in Y$ and the A_i s are disjoint.
- 5) aRc ($a, c \in X$ and def. of R).

Equivalence Relations and Partitions

Proof (continuation).

$$R = \{ (a, b) \mid a, b \in A_i \}.$$

- Transitivity

- 1) aRb and bRc .
- 2) Exists $X \in \{ A_i \mid i \in I \}$ such that $a, b \in X$ by definition of R .
- 3) Exists $Y \in \{ A_i \mid i \in I \}$ such that $b, c \in Y$ by definition of R .
- 4) $X = Y$ because $b \in X$ and $b \in Y$ and the A_i s are disjoint.
- 5) aRc ($a, c \in X$ and def. of R).

Now, $[a]_R = \{ s \mid (a, s) \in R \}$ and by the definition of the relation R , these equivalence classes correspond to the sets A_i . ■

References

- Kenneth H. Rosen [1988] (2004). *Matemática Discreta y sus Aplicaciones*. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).
- [1988] (2012). *Discrete Mathematics and Its Applications*. 7th ed. McGraw-Hill (cit. on pp. 3, 29).